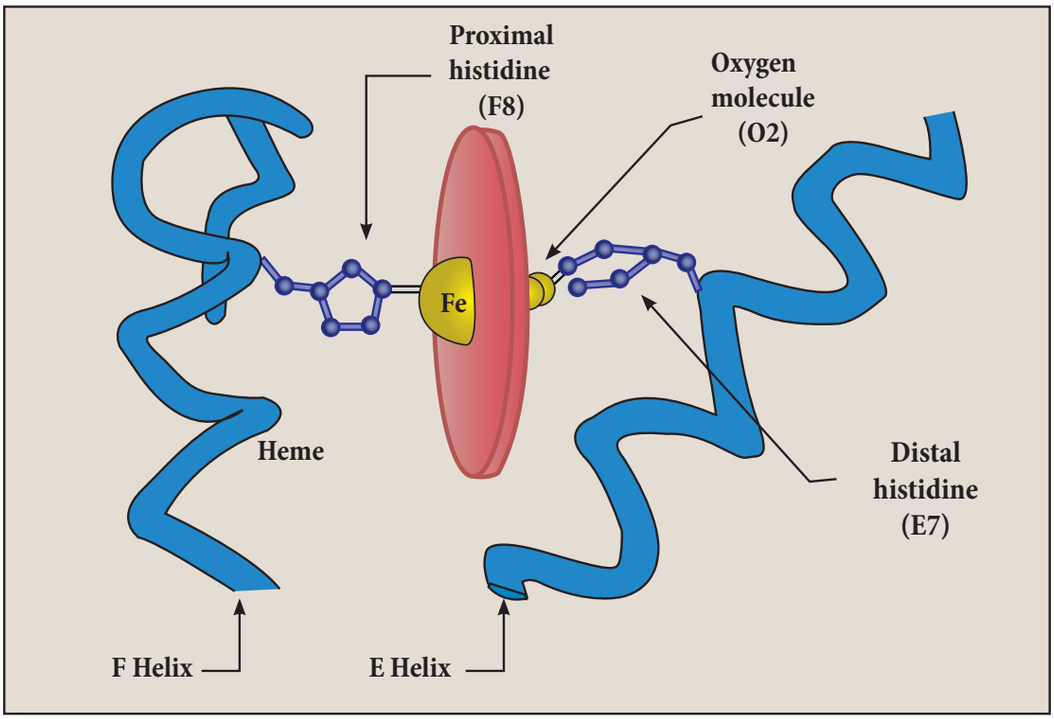


Figure 3.14 Binding of oxygen to heme in hemoglobin

Haemoglobin is an alpha helical protein, meaning it does not contain β sheets as its secondary structural elements. The haemoglobin tetramer structure could be considered as a dimerized dimer of (αβ)1 and (αβ)2. The α and β chain in each dimer are held together strongly by hydrophobic interactions. The interactions between (αβ)1 and (αβ)2 are comparatively weaker hydrogen bonds and ionic interactions. This allows the dimers to move with respect to each other forming two different conformational states: a relaxed 'R' conformation and a taut 'T' conformation. The binding and release of oxygen switches the hemoglobin between these two states.

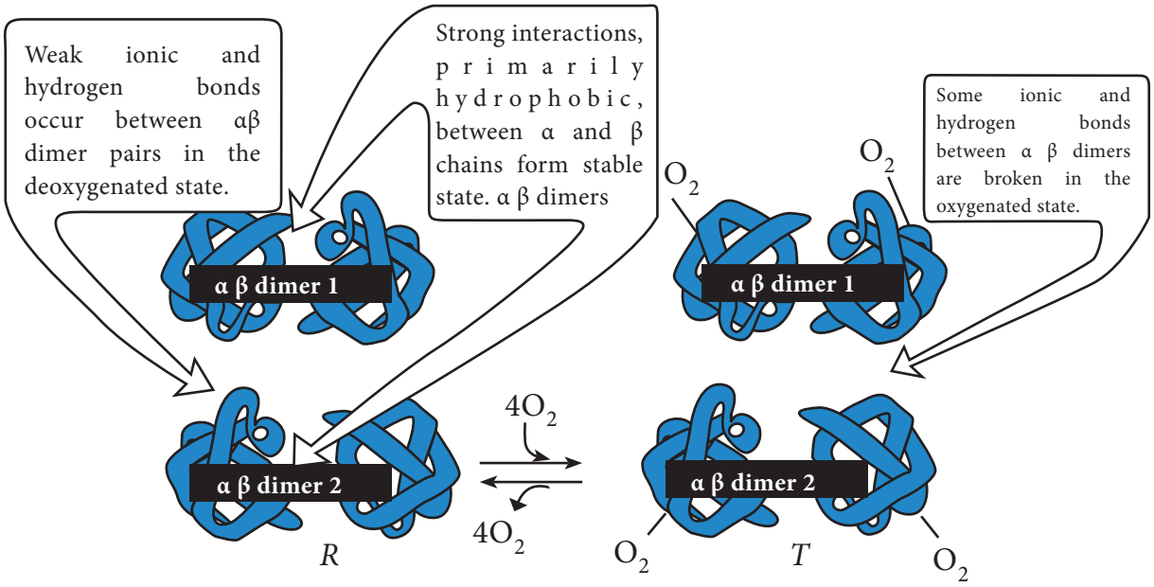


Figure 3.15 R and T conformations of haemoglobin

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